

Gaurav Singh

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Research profile

Researcher and educator working across **human-centred design, experience design**, software engineering and **applied machine learning**, with a background spanning engineering and design schools. Published research in HCI, visual-mathematical literacy, auxetic structures and transport modelling. Headed the M.Des Design Computation programme and set up the Experimental Maths Lab at Srishti Manipal Institute. Maintain Mathscapes as an independent mathematical practice.

Research methods and technical skills

Design and HCI: human-centred design, experience design, interaction design and physical computing.

Generative time-series modelling: Python, PyTorch and PyTorch Lightning for synthetic AQI data generation and transformer-based modelling.

Simulation-based machine learning: finite-element simulation data, K-means image labelling and logistic regression for deformation-mode detection.

Representation and feature design: visual representations, feature and label construction, and analysis of how representation choices shape model learning.

Education

2023-2025	Master of Information Technology Griffith University, Gold Coast, Australia; GPA 6.31/7 Thesis: "Enhancing Climate Forecasting with Advanced Generative Models for Synthetic AQI Data", supervised by Dr. Elizabeth Chang.
2013-2015	Advanced Diploma, Innovation & Experience Design <u>Srishti Institute of Art, Design and Technology, Bangalore, India</u>
2008-2012	B.Tech, Computer Science & Engineering JNTU Anantapur, India; First Class with Distinction

Publications

A Framework for Visual-Mathematical Literacy in Applied Machine Learning: Why Representation Choices Shape What Models Can Learn. G. Singh, R. S. Dhari. Machine Learning: Engineering, 2026. <https://doi.org/10.1088/3049-4761/ae7df3>

Techno-Economic Pathways Modeling and Nonlinear Optimized SEEA-ROI Longitudinal Dynamic Simulation for Decarbonizing Australian Heavy Transportation Systems. G. Singh, E. Chang, Y. Karaca. Fractals, 2026. <https://doi.org/10.1142/S0218348X26400633>

Automated detection of deformation mechanisms in re-entrant honeycomb auxetics using machine learning. G. Singh, R. S. Dhari, Z. Javanbakht. International Journal of Protective Structures, 2025. <https://doi.org/10.1177/20414196241281069>

ReRide: A Bike Area Network for Embodied Self-monitoring during Motorbike Commute. N. Bagalkot, G. Singh, V. Rath, T. Sokoler, A. Shukla. TEI 2019, 443-450, 2019. <https://doi.org/10.1145/3294109.3300986>

ReRide. N. Bagalkot, T. Sokoler, R. Shaikh, G. Singh, A. E. Lillie, P. Dixit, A. Rai, V. Chakravarthy, A. Senthil. INTERACT 2017, LNCS 10516, 2017. https://doi.org/10.1007/978-3-319-68059-0_43

SnapTag: Leveraging Situated Memory to enhance self-efficacy for well-being. S. Baadkar, G. Singh, A. Saraf, N. Bagalkot. India HCI 2014, 136-141, 2014. <https://doi.org/10.1145/2676702.2676719>

Academic appointment and leadership

- 2017-2022 **Head of Programme, M.Des Design Computation**, Srishti Manipal Institute, Bangalore, India
- Headed the Master of Design in Design Computation.
 - Set up the Experimental Maths Lab.
 - Taught interaction design and human-computer interaction alongside programming, algorithms, machine learning and physical computing.

Teaching and curriculum

Designed and taught **30 courses and workshops** (2013-2022), primarily at Srishti Manipal Institute in Bangalore, connecting interaction design and human-computer interaction with creative coding, algorithmic botany, physical computing and machine learning.

Designed and facilitated Deconstructing Algorithms: a thesis studio on algorithmic literacy in 2020. Supervised five undergraduate students investigating explainable AI and impact-investment information.

Thesis supervision

- 2020 Discovering Algorithmic Literacy through Explainable AI
Thesis supervisor; students: D. Dixit, S. Singh
- 2020 Financial indicators that motivate impact investment
Thesis supervisor; students: A. Avadhana, N. Patny, S. Mishra
- 2013 Location and arrival prediction
Thesis supervisor; students: Shobha R., Pavani A., Kalpana, JNTU Anantapur

Academic service

- 2018 Demonstrations track co-chair, IndiaHCI 2018
Bangalore, India
- 2017 Cumulus Conference, organising team
Bangalore, India

Awards

- 2024 Griffith Award for Academic Excellence (Griffith University)
- 2024 STEM Professional Gold Achievement (Griffith University)
- 2017 Topcoder Open Design Champion (India Regionals)
- 2012 Invited to Topcoder Open Studio Finals (6th, Orlando)
- 2011 Invited to Topcoder Open Studio Finals (11th, Fort Lauderdale)

Grants

- 2023 Adobe Fund for Design (Design grant)
- 2021 Adobe Fund for Design (Design grant)

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